CMSC433 / Part 01

Type foundations

Topic mock

Topic mock, using the sample's question styles. Attempt independently; feedback after the whole file. Untimed practice, no points assigned. Current exam scope and rules are unconfirmed.

For each independent expression, give its most general type, including necessary constraints, or write "ill-typed" and explain the conflict. Show enough reasoning to justify your answer. These are type questions, not evaluation questions.

(a) `\p -> (snd p, fst p)`

Answer: [Type and reasoning.]

(b) `\xs -> [x | x <- xs, x]`

Answer: [Type and reasoning.]

(c) `\xs -> map show xs`

Answer: [Type and reasoning.]

(d) `\f x -> (f x, f x)`

Answer: [Type and reasoning.]

(e) `\x -> x x`

Answer: [Type or conflict.]

(f) `\x -> (x + 1, x == x)`

Answer: [Type and reasoning.]

Supplied signatures

```
fst :: (a,b) -> a
snd :: (a,b) -> b
map :: (a -> b) -> [a] -> [b]
show :: Show a => a -> String
(+) :: Num a => a -> a -> a
(==) :: Eq a => a -> a -> Bool
```